

SYNTHESIS PROJECT

# An AprilTag-Based Approach to Trajectory Recording and Relocalization for Mobile Robots

**Joao Pedro Martins do Lago Reis and Yann Kelvem da Silva Ramos**

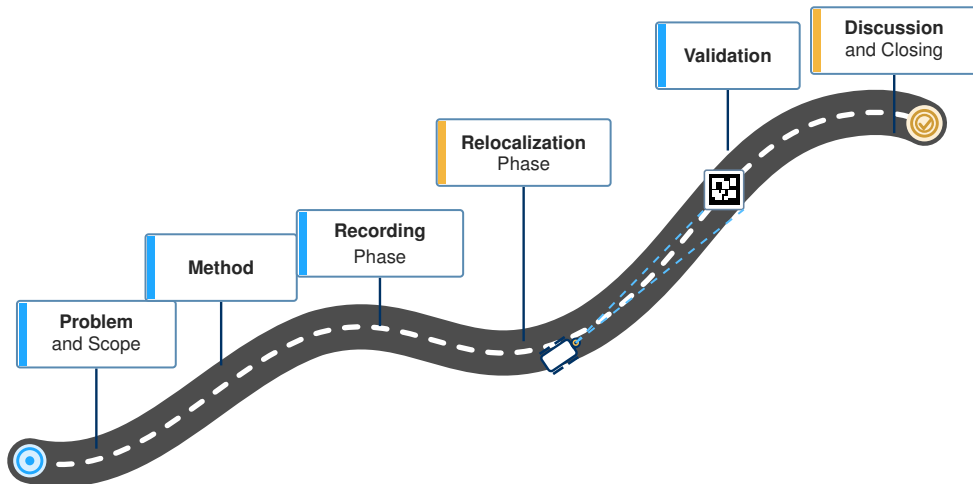
Master's degree in artificial perception and robotics

March 02, 2026

**Supervisor:** Dr. Sébastien Lengagne

**Evaluator:** Dr. Aufrère Romuald

# Outline



# Problem and Scope

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## For whom

Polytech practical-work students using  
AgileX LIMO in indoor AprilTag setups.

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## Why

Need a reproducible workflow to record a path and recover pose relative to it for control-law exercises.

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Recording alone is insufficient because odometry drift breaks trajectory consistency across runs.

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## Problem

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## Scope

In: offline reconstruction and relocalization outputs ( $D$ ,  $\phi$ ).

Out: control-law and trajectory replay.

# Method

## Three-Stage Method



**1) Record**  
detections dataset

## Main Equation

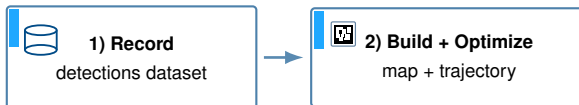
Recording

$${}^{\text{cam}}T_{\text{tag}}(t)$$

## Artifacts

# Method

## Three-Stage Method



## Main Equation

**Build + Optimize**

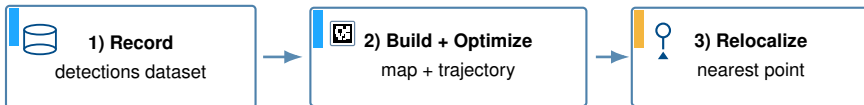
$$\min_{\{w_{T_{tag_i}}\}} \sum_{(i,j) \in \mathcal{E}} \rho(\|err_{ij}\|^2)$$

## Artifacts



# Method

## Three-Stage Method



## Main Equation

### Relocalization

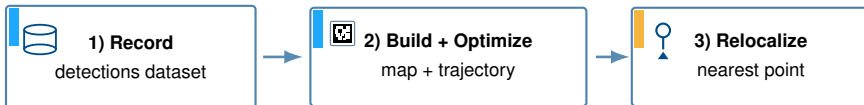
$$D = \|\mathbf{p}_{\text{cur}} - \mathbf{p}_{\text{nearest}}\|$$

$$\phi = \text{wrap}(\psi_{\text{cur}} - \psi_{\text{nearest}})$$

## Artifacts

# Method

## Three-Stage Method



## Main Equation

### Relocalization

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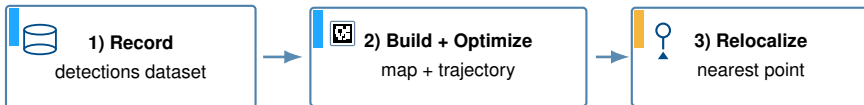
## Artifacts



tag\_map.yaml

# Method

## Three-Stage Method



## Main Equation

### Relocalization

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## Artifacts



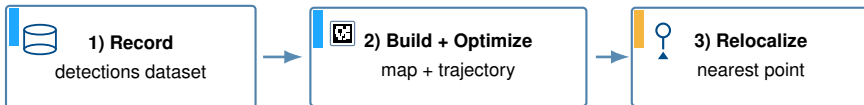
tag\_map.yaml



trajetoria\_camera.csv

# Method

## Three-Stage Method



## Main Equation

### Relocalization

$$D = \|\mathbf{p}_{\text{cur}} - \mathbf{p}_{\text{nearest}}\|$$

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## Artifacts



tag\_map.yaml



trajectoria\_camera.csv



$D$  [m],  $\phi$  [deg]

# Recording Phase

## Recording Phase — Template

### To fill later:

- ▶ Inputs and sensors
- ▶ Offline optimization steps
- ▶ Generated files
- ▶ Key interpretation points

# Relocalization Phase

## Relocalization Phase — Template

### To fill later:

- ▶ Online inputs
- ▶ Nearest-path logic
- ▶ Distance/heading outputs
- ▶ RViz visualization items

# Validation

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## Validation — Template

### To fill later:

- ▶ Real scenarios
- ▶ Simulation scenario
- ▶ Quantitative metrics
- ▶ Comparison criteria

# Discussion

## Discussion and Conclusion — Template

### To fill later:

- ▶ Main findings
- ▶ Current limitations
- ▶ Scope boundaries
- ▶ Next steps



# Conclusion

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# Thank you for listening!

Any questions?

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Joao Pedro Martins do Lago Reis | Yann Kelvem da Silva Ramos  
Record, Map, and Relocalize with AprilTags